TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Efficient Inference Framework of Large-Scale AI Models on Accelerator-Centric Architectures			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	System Integration	Semiconductor Devices and Systems	Other Semiconductor Devices
Concept of Proposed Technology	We propose an accelerator-first serving framework that enables efficient inference of large language models on specialized data center inference accelerators. The framework focuses on two dominant bottlenecks in production serving: (i) memory pressure and bandwidth from growing key-value caches in long-context decoding, and (ii) expert management overhead in mixture-of-experts models where only a subset of experts is activated per token. The main function is a framework that provides: a) a hardware-adaptation layer for AI accelerator cards; b) a cache manager with selective token access to avoid unnecessary loading from long contexts; c) an expert manager that classifies frequently used experts and places them in the most appropriate memory tier; and d) a serving runtime that performs batching, routing-aware scheduling, and elastic resource allocation for multi-tenant development.			
Rationale for Joint R&D Project	Long-context serving makes inference increasingly memory-bound. During decoding, each token attends to prior tokens using a growing key-value cache. For a representative 7B-class model (32 layers, 32 heads, 128 head dimension) stored in 16-bit precision, the total key-value cache footprint			

	is approximately 512 KB/token. At 128k tokens, this is ~64 GB of key-value cache, which can exceed or saturate the high-bandwidth memory of a single accelerator and drive frequent memory traffic even when only a small subset of history is relevant [1][2]. In parallel, mixture-of-experts models introduce a different memory bottleneck: expert weights may not fit in fast device memory, and routing patterns are skewed so that a small subset of experts is frequently activated. Without an expert-aware memory manager, expert loads, swaps, and synchronization dominate latency and reduce utilization [3][4][5]. **Need for collaboration.** An accelerator-serving framework is inherently co-designed across hardware, compiler, runtime, and model software. Academic and research partners can develop new algorithms for dynamic key-value retrieval and expert management, define fair benchmarks, and validate accuracy/quality tradeoffs. However, achieving production-grade performance requires tight integration with accelerator software stacks (operators, quantization formats, kernels, compiler passes, and performance counters) [6], as well as access to real hardware platforms and deployment constraints [7][8]. Korean small and medium sized enterprises are uniquely positioned to provide these assets and to translate the results into commercial serving products.
Key Words	1. Serving framework 2. AI accelerator 3. Long-context processing 4. Mixture-of-experts model

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• Accelerator-oriented serving runtime and backend interface • Dynamic key-value cache indexing and retrieval for long contexts • Key-value cache memory tiering, compression, and eviction • Expert management for mixture-of-experts inference • System-level scheduling and resource allocation			
R&D Target Specifications	Targets are defined relative to a reference baseline on the same accelerator platform: vendor-provided runtime with dense attention (full key-value scan) and static mixture-of-experts weight placement. Performance will be reported for representative long-context and mixture-of-experts workloads agreed with industry partners. Performance targets: 1) memory read traffic with 40% reduction at 64k context and >= 60% reduction at 128k context vs. dense attention baseline; 2) per-token latency with 30% to 50% reduction vs. baseline; 3) > 256K context on high-bandwidth memory class accelerators; 4) throughput for MoE models with >= 25% improvement with equal model quality and batch size.			
R&D Plan	We will follow an iterative co-design methodology: baseline port and measurement on partner accelerators, algorithm development informed by real bottlenecks, tight integration with accelerator kernels and compiler toolchains, and staged delivery of production-ready components through repeated end-to-end evaluations. • **Task 1 - Baseline serving stack and benchmarking:** Port a representative large language model serving baseline to partner accelerators, define workloads (long-context and mixture-of-experts), and establish a reproducible benchmark suite with latency, throughput, bandwidth, and power metrics. (Month 1-6) • **Task 2 - Accelerator backend development:** Implement the backend interface, operator coverage, and kernels required for attention, cache operations, and mixture-of-experts routing. Integrate quantization formats supported by each accelerator and validate numerical correctness. (Month 7-12) • **Task 3 - Dynamic key-value cache retrieval:** Implement cluster-based key-value layout, centroid/index build, query-to-cluster matching, selective fetch, and adaptive sparsity control. Validate quality vs. dense attention and tune budgets to service-level objectives. (Month 13-20) • **Task 4 - Cache tiering and capacity scaling:** Add hot/cold cache tiering across on-chip memory and high-bandwidth memory, optional cold-cache compression/quantization, and eviction policies for multi-tenant serving. Stress test at 256k+ contexts. (Month 21-24) • **Task 5 - Hot/cold expert management for mixture-of-experts:** Build expert popularity tracking, placement and replication policies, cold expert streaming, and routing-aware batching. Optimize expert paging and validate throughput/latency improvements. (Month 25-30) • **Task 6 - End-to-end serving integration and prototyping:** Integrate all components into a deployable serving framework, validate on multi-card systems, and deliver partner-specific proofs of concept with documentation and technology transfer. (Month 31-36)			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will lead the algorithmic and system software innovations and provide the neutral evaluation and benchmarking foundation needed for credible commercialization by participating Korean small and medium sized enterprises. • **Technical contributions:** Design and implement dynamic key-value cache retrieval algorithms (cluster formation, indexing, query-to-cluster matching, adaptive sparsity control) and validate quality impacts. Design expert management algorithms for mixture-of-experts models (hot/cold expert classification, placement, replication, and prefetch policies). Architect the modular serving runtime, including scheduler, batching, and memory manager, with a clean backend interface for accelerator integration. Develop a correctness and regression test harness to ensure functional parity across accelerator backends. Create a reproducible benchmark suite and evaluation methodology (latency percentiles, tokens/s, memory bandwidth, and energy). • **Resources to be provided**: Dedicated research and engineering personnel for algorithm development, runtime implementation, and evaluation. Development infrastructure (servers, storage, and continuous integration) for building and testing the framework; reference GPU systems for cross-validation when needed. Documentation, integration guides, and training sessions for partner engineering teams to accelerate adoption and productization. • **Deliverables to partners:** Reference implementation of dynamic key-value cache manager and expert manager integrated into the serving runtime. Performance reports and tuning recommendations tailored to each partner accelerator platform. Technology transfer package: APIs, deployment scripts, and partner-facing documentation suitable for incorporation into commercial designs.
Expected R&R of Korean SME Partner	Korean small and medium sized enterprise partners will provide the accelerator-specific software and hardware environment required to translate research innovations into deployable and product-ready serving solutions. SME partners are expected to provide access to accelerator hardware (cards and/or servers), software development kits, compiler toolchains, and runtime libraries needed for development and profiling. Co-develop and optimize accelerator kernels for attention operations, key-value cache storage, and mixture-of-experts routing, including low-precision formats and memory primitives. Expose and document performance counters, profiling tools, and recommended optimization practices for the accelerator platform. Support integration into partner serving stacks (drivers or container images) and ensure compatibility with deployment environments.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Through this joint effort, Korean SMEs on AI accelerators can be paired with a differentiated serving framework that specifically targets the hardest inference regimes (long context and mixture-of-experts), strengthening the competitiveness of Korean small and medium sized enterprises in the global inference market.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization